

HKCEE 1993 Mathematics II

Label 1 If $f(x) = 10^{2x}$, then $f(4y) =$

- A. 10^{4y}
- B. 10^{2+4y}
- C. 10^{8y}
- D. 40^y
- E. 40^{2y}

Label 2 If $s = \frac{n}{2} [2a + (n-1)d]$, then $d =$

- A. $\frac{2(s-an)}{n(n-1)}$
- B. $\frac{2(s-an)}{(n-1)}$
- C. $\frac{s}{n(n-1)}$
- D. $\frac{as-n}{a(n-1)}$
- E. $\frac{4(s-an)}{n(n-1)}$

Label 3 Simplify $(x^2 - \sqrt{3}x + 1)(x^2 + \sqrt{3}x + 1)$.

- A. $x^4 + 1$
- B. $x^4 - x^2 + 1$
- C. $x^4 + x^2 + 1$
- D. $x^4 - 3x^2 - 2\sqrt{3}x - 1$
- E. $x^4 + 3x^2 - 2\sqrt{3}x^2 + \sqrt{3}x - 1$

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4. $\frac{\sqrt{a}}{\sqrt{a}-\sqrt{b}} + \frac{\sqrt{a}}{\sqrt{a}+\sqrt{b}}$

- A. $\frac{1}{a+2\sqrt{ab}-b}$
- B. $\frac{1}{a-b}$

Figure 3b Density 514

C. $\frac{\sqrt{b+\sqrt{a}}}{2\sqrt{a}}$

D. $\frac{b+2\sqrt{ab}-a}{a-b}$

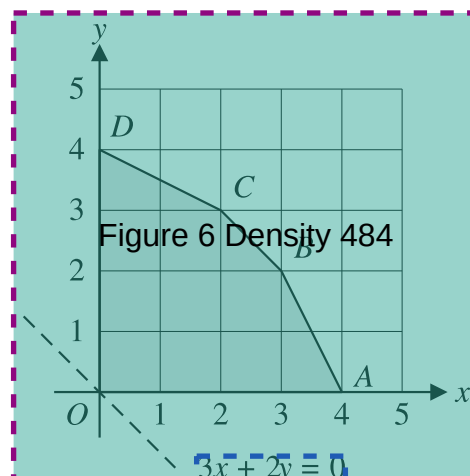
Figure 3a Density 651

E. $\frac{a+b}{a-b}$

Label 5 If $3x^2 + ax - 5 \equiv (bx - 1)(2 - x) - 3$, then

- A. $a = -5, b = -3$
- B. $a = -5, b = 3$
- C. $a = -3, b = -5$
- D. $a = 5, b = -3$
- E. $a = 3, b = 5$

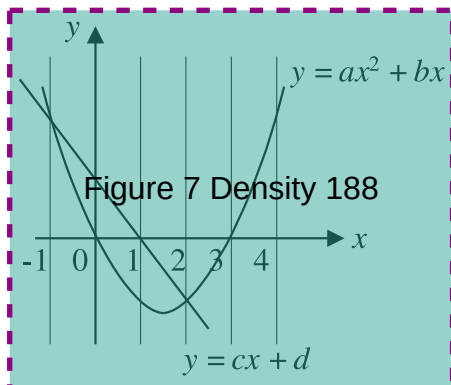
Label 6



Find the greatest value of $3x + 2y$ if (x, y) is a point lying in the region $OABCD$ (including the boundary).

- A. 15
- B. 13
- C. 12
- D. 9
- E. 8

Label 7



The diagram shows the graphs of $y = ax^2 + bx$ and $y = cx + d$. The solutions of the equation $ax^2 + bx = cx + d$ are

- A. $-1, 1$
- B. $-1, 2$
- C. $0, 1$
- D. $0, 3$
- E. $1, 3$

Label 8 If $\log(p + q) = \log p + \log q$, then

- A. $p = q = 1$
- B. $p = \frac{q}{q-1}$
- C. $p = \frac{q}{q+1}$
- D. $p = \frac{q+1}{q}$
- E. $p = \frac{q-1}{q}$

Label 9 The expression $x^2 - 2x + k$ is divisible by $(x + 1)$. Find the remainder when it is divided by $(x + 3)$.

- A. 1
- B. 4
- C. 12
- D. 16
- E. 18

Label 10 If $3, a, b, c, 23$ are in A.P., then

$a + b + c =$

- A. 13
- B. 26
- C. 33
- D. 39
- E. 65

Label 11 Find the H.C.F. and L.C.M. of ab^2c and abc^3

- | | H.C.F. | L.C.M. |
|----|-------------|-------------|
| A. | a | $a^2b^3c^4$ |
| B. | abc | ab^2c^3 |
| C. | abc | $a^2b^3c^4$ |
| D. | ab^2c^3 | aba |
| E. | $a^2b^3c^4$ | abc |

Label 12 If α and β are the roots of the quadratic equation $x^2 - 3x - 1 = 0$, find the value of $\frac{1}{\alpha} + \frac{1}{\beta}$.

- A. -3
- B. -1
- C. $\frac{1}{3}$
- D. $\frac{2}{3}$
- E. 3

Label 13 If the simultaneous equations $\begin{cases} y = x^2 - k \\ y = x \end{cases}$ have only one solution, find k .

- A. -1
- B. 1
- C. -4
- D. 4
- E. 1

Label 14

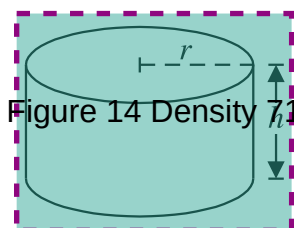


Figure 14 Density 71

The price of a cylindrical cake of radius r and height h varies directly as the volume. If $r = 5$ cm and $h = 4$ cm, the price is \$30. Find the price when $r = 4$ cm and $h = 6$ cm.

- A. \$25
- B. \$28.80
- C. \$31.50
- D. \$36
- E. \$54

Label 15

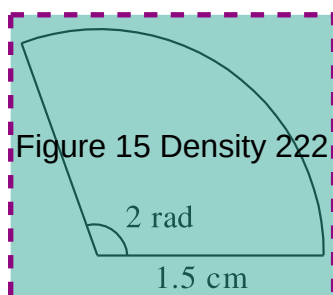


Figure 15 Density 222

Find the perimeter of the sector in the figure.

- A. 2.25 cm
- B. 3 cm
- C. $\left(\frac{\pi}{60} + 3\right)$ cm
- D. 4.5 cm
- E. 6 cm

Label 16

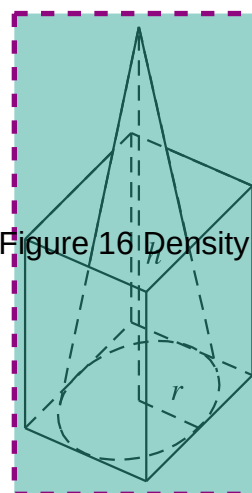


Figure 16 Density 0

In the figure, the base of the conical vessel is inscribed in the bottom of the cubical box. If the box and the conical vessel have the same capacity, find h .

- A. $24 : \pi$
- B. $3 : 1$
- C. $6 : \pi$
- D. $3 : \pi$
- E. $8 : 3\pi$

Label 17

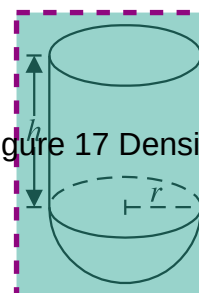


Figure 17 Density 0

The figure shows a solid consisting of a cylinder of height h and a hemisphere of radius r . The area of the curved surface of the cylinder is twice that of the hemisphere. Find the ratio volume of cylinder : volume of hemisphere.

- A. $1 : 3$
- B. $2 : 3$
- C. $3 : 4$
- D. $3 : 2$
- E. $3 : 1$

Label 18
18. A merchant marks his goods 25% above the cost. He allows 10 % discount on the marked price for a cash sale. Find the percentage profit the merchant makes for a cash sale.

- A. 12.5%
- B. 15%
- C. 22.5%
- D. 35%
- E. 37.5%

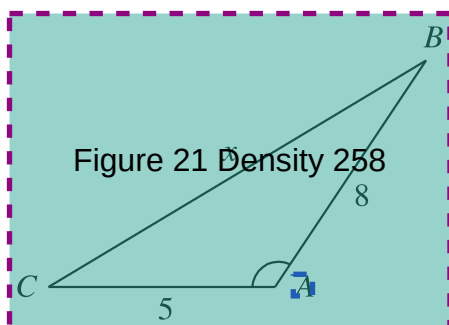
Label 19
19. $\frac{\cos \theta}{1 - \sin^2 \theta} \cdot \frac{1 - \cos^2 \theta}{\sin \theta} =$

- A. $\sin \theta$
- B. $\cos \theta$
- C. $\tan \theta$
- D. $\frac{1}{\sin \theta}$
- E. $\frac{1}{\cos \theta}$

Label 20
20. $\cos^4 \theta - \sin^4 \theta + 2 \sin^2 \theta =$

- A. 0
- B. 1
- C. $(1 - \sin^2 \theta)^2$
- D. $(1 - \cos^2 \theta)^2$
- E. $(\cos^2 \theta - \sin^2 \theta)^2$

Label 21
21.



In the figure, $\cos A = -\frac{4}{5}$. Find a .

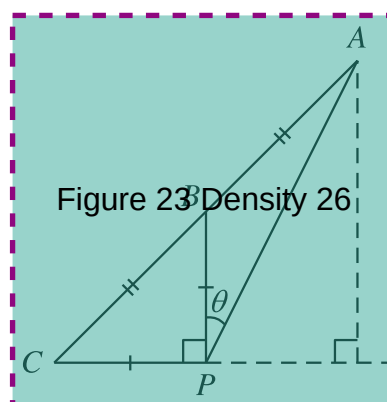
- A. $\sqrt{153}$
- B. $\sqrt{137}$

- C. $\sqrt{89}$
- D. $\sqrt{41}$
- E. $\sqrt{25}$

Label 22
22. The largest value of $3\sin^2 \theta + 2\cos^2 \theta - 1$ is

- A. 1
- B. $\frac{3}{2}$
- C. 2
- D. 3
- E. 4

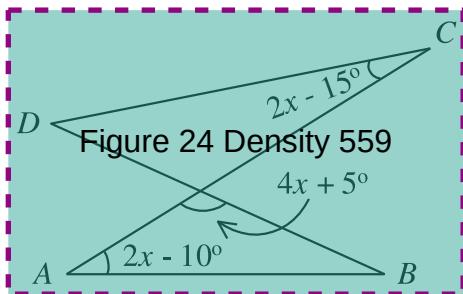
Label 23
23.



In the figure, $AB = BC$, $BP = CP$ and $BP \perp CP$. Find $\tan \theta$.

- A. $\frac{1}{4}$
- B. $\frac{1}{3}$
- C. $\frac{1}{2}$
- D. $\frac{1}{\sqrt{3}}$
- E. $\frac{\sqrt{3}}{2}$

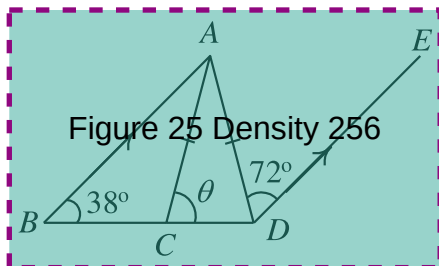
Label 24
24



In the figure, points A , B , C and D are concyclic. Find x .

- A. 20°
- B. 22.5°
- C. 25°
- D. 27.5°
- E. 30°

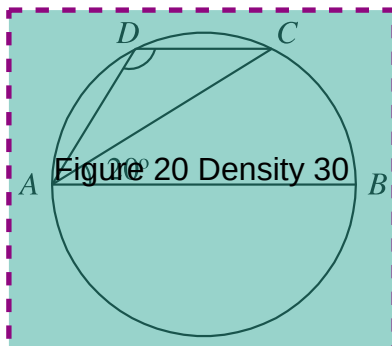
Label 25
25



In the figure, $BA \parallel DE$ and $AC = AD$. Find θ .

- A. 34°
- B. 54°
- C. 70°
- D. 72°
- E. 76°

Label 26
26



In the figure, AB is a diameter. Find $\angle ADC$.

- A. 100°

- B. 110°
- C. 120°
- D. 135°
- E. 140°

Label 27
27 If the points $(1, 1)$, $(3, 2)$ and $(7, k)$ are on the same straight line, then $k =$

- A. 3.
- B. 4.
- C. 6.
- D. 7.
- E. 10.

Label 28
28 $A(0, 0)$, $B(5, 0)$ and $C(2, 6)$ are the vertices of a triangle. $P(9, 5)$, $Q(6, 6)$ and $R(2, -9)$ are three points. Which of the following triangles has/have area(s) greater than the area of $\triangle ABC$?

- I. $\triangle ABP$
- II. $\triangle ABQ$
- III. $\triangle ABR$

- A. I only
- B. II only
- C. III only
- D. I and II only
- E. II and III only

Label 29
29 A circle of radius 1 touches both the positive x -axis and the positive y -axis. Which of the following is/are true?

- I. Its centre is in the first quadrant.
- II. Its centre lies on the line $x - y = 0$.
- III. Its centre lies on the line $x + y = 1$.

- A. I only
- B. II only
- C. III only
- D. I and II only
- E. I and III only

Label 30
30 What is the area of the circle $x^2 + y^2 - 10x + 6y - 2 = 0$?

- A. 32π

- B. 64π
 C. 36π
 D. 134π
 E. 138π

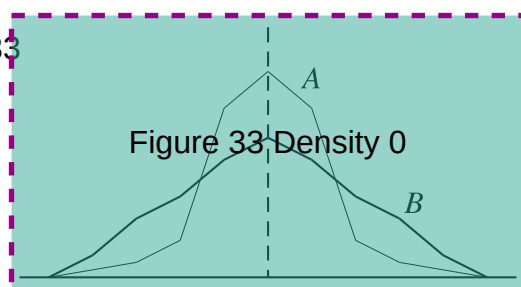
Label 31
 31. Two fair dice are thrown. What is the probability of getting a total of 5 or 10?

- A. $\frac{1}{9}$
 B. $\frac{5}{36}$
 C. $\frac{1}{6}$
 D. $\frac{7}{36}$
 E. $\frac{2}{9}$

Label 32
 32. A group of n numbers has mean m . If the numbers 1, 2 and 6 are removed from the group, the mean of the remaining $n - 3$ numbers remains unchanged. Find m .

- A. 1
 B. 2
 C. 3
 D. 6
 E. $n = 3$

Label 33
 33.



The figure shows the frequency polygons of two symmetric distributions A and B with the same mean. Which of the following is/are true?

- I. Interquartile range of A
 II. Interquartile range of B
 III. Standard deviation of A

Standard deviation of B
 III. Mode of A > Mode of B

- A. I only
 B. II only
 C. III only
 D. I and II only
 E. II and III only

Label 34
 34. If $9^{x+2} = 36$, then $3^x =$

- A. $\frac{2}{3}$
 B. $\frac{4}{3}$
 C. 2
 D. $\sqrt{6}$
 E. 9

Label 35
 35. If $a : b = 2 : 3$ and $b : c = 5 : 3$, then $\frac{a+b+c}{a-b+c} =$

- A. -2
 B. $\frac{5}{2}$
 C. 4
 D. $\frac{17}{2}$
 E. 31

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 36.

x	Sign of $f(x)$
3.56	
3.58	-
3.57	+
3.575	+

From the table, a root of the equation $f(x) = 0$ is

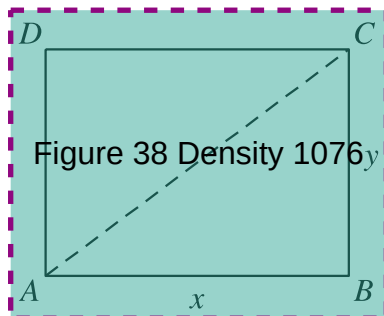
- A. 3.57 (correct to 3 sig. fig.).
 B. 3.575 (correct to 4 sig. fig.).
 C. 3.5775 (correct to 5 sig. fig.).
 D. 3.5725 (correct to 4 sig. fig.).
 E. 3.58 (correct to 3 sig. fig.).

Label 37
37 Given that the positive numbers p, q, r, s are in G.P., which of the following must be true?

- I. kp, kq, kr, ks are in G.P., where k is a non-zero constant.
- II. a^p, a^q, a^r, a^s are in G.P., where a is a positive constant.
- III. $\log p, \log q, \log r, \log s$ are in A.P.

- A. I only
- B. II only
- C. I and II only
- D. I and III only
- E. I, II and III only

Label 38
38



In the figure, the rectangle has perimeter 16 cm and area 15 cm^2 . Find the length of its diagonal AC.

- A. $\sqrt{32}$ cm
- B. $\sqrt{34}$ cm
- C. 7 cm
- D. $\sqrt{226}$ cm
- E. $\sqrt{241}$ cm

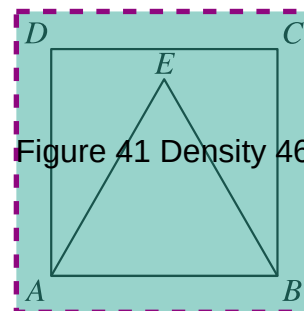
Label 39
39 In factorizing the expression $a^4 + a^2b^2 + b^4$, we find that

- A. $(a^2 - b^2)$ is a factor.
- B. $(a^2 + b^2)$ is a factor.
- C. $(a^2 - ab - b^2)$ is a factor.
- D. $(a^2 - ab + b^2)$ is a factor.
- E. it cannot be factorized.

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40 If the solution of the inequality $x^2 - ax + 6 \leq 0$ is $c \leq x \leq 3$, then

- A. $a = 5, c = 2$
- B. $a = -5, c = 2$
- C. $a = 5, c = -2$
- D. $a = 1, c = -2$
- E. $a = -1, c = 2$

Label 41
41

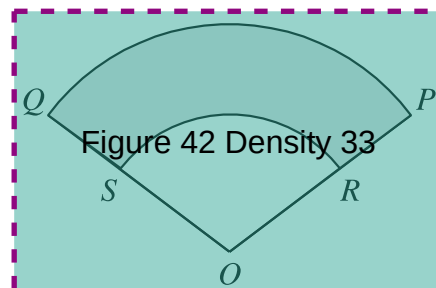


In the figure, ABCD is a square and ABE is an equilateral triangle.

$$\frac{\text{Area of } ABE}{\text{Area of } ABCD} =$$

- A. $\frac{1}{4}$
- B. $\frac{1}{3}$
- C. $\frac{\sqrt{3}}{8}$
- D. $\frac{\sqrt{3}}{4}$
- E. $\frac{\sqrt{3}}{2}$

Label 42
42



In the figure, the radii of the sectors OPQ and ORS are 5 cm and 3 cm respectively. $\frac{\text{Area of shaded region}}{\text{Area of sector OPQ}} =$

- A. $\frac{4}{25}$
 B. $\frac{2}{5}$
 C. $\frac{9}{25}$
 D. $\frac{16}{25}$
 E. $\frac{21}{25}$

Label 43
 43. Which of the following gives the compound interest on \$ 10 000 at 6% p.a. for one year, compounded monthly?

- A. $\$ 10\,000 \times \frac{0.06}{12} \times 12$
 B. $\$ 10\,000(1.06^{12} - 1)$
 C. $\$ 10\,000 \left(1 + \frac{0.06}{12}\right)^{12}$
 D. $\$ 10\,000 \left[\left(1 + \frac{0.06}{12}\right)^{12} - 1\right]$
 E. $\$ 10\,000 \left[\left(1 + \frac{0.6}{12}\right)^{12} + 1\right]$

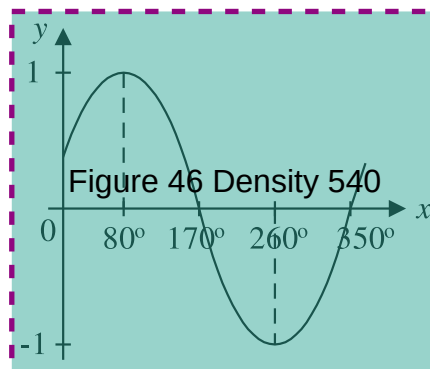
Label 44
 44. Originally $\frac{2}{3}$ of the students in a class failed in an examination. After taking a re-examination, 40% of the failed students passed. Find the total pass percentage of the class.

- A. $26\frac{2}{3}\%$
 B. $33\frac{1}{3}\%$
 C. 40%
 D. 60%
 E. $73\frac{1}{3}\%$

Label 45
 45. Solve $\tan^4 \theta + 2\tan^2 \theta - 3 = 0$ for $0^\circ \leq \theta < 360^\circ$.

- A. $45^\circ, 135^\circ$ only
 B. $45^\circ, 225^\circ$ only
 C. $45^\circ, 60^\circ, 225^\circ, 240^\circ$
 D. $45^\circ, 120^\circ, 225^\circ, 300^\circ$
 E. $45^\circ, 135^\circ, 225^\circ, 315^\circ$

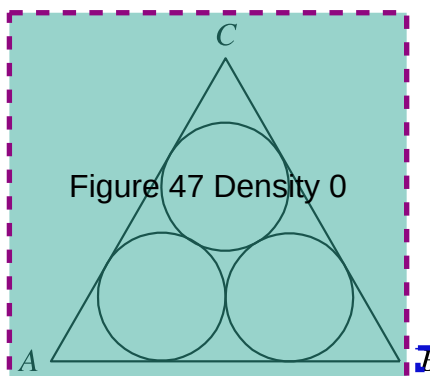
Label 46
 46.



The figure shows the graph of the function

- A. $y = \sin(350^\circ - x)$
 B. $y = \sin(x + 10^\circ)$
 C. $y = \cos(x + 10^\circ)$
 D. $y = \sin(x - 10^\circ)$
 E. $y = \cos(x - 10^\circ)$

Label 47
 47.

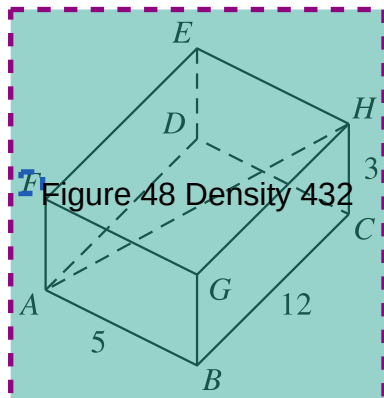


In the figure, ABC is an equilateral triangle and the radii of the three circles are each equal to 1. Find the perimeter of the triangle.

- A. 12
 B. $3(1 + \tan 30^\circ)$
 C. $6(1 + \tan 30^\circ)$

- D. $3\left(1 + \frac{1}{\tan 30^\circ}\right)$
 E. $6\left(1 + \frac{1}{\tan 30^\circ}\right)$

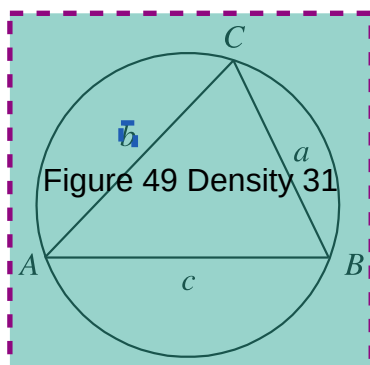
Label 48



In the figure, $ABCDEFGH$ is a cuboid. The diagonal AH makes an angle θ with the base $ABCD$. Find $\tan \theta$.

- A. $\frac{3}{5}$
 B. $\frac{3}{12}$
 C. $\frac{3}{13}$
 D. $\frac{3}{\sqrt{178}}$
 E. $\frac{\sqrt{153}}{5}$

Label 49



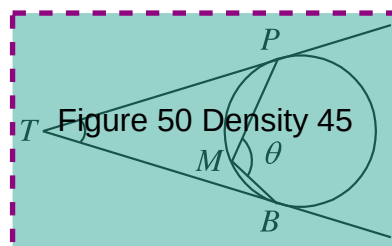
In the figure, if $\text{arc } BC : \text{arc } CA : \text{arc } AB = 1 : 2 : 3$, which of the following is/are true?

- I. $\angle A : \angle B : \angle C = 1 : 2 : 3$

- II. $a : b : c = 1 : 2 : 3$
 III. $\sin A : \sin B : \sin C = 1 : 2 : 3$

- A. I only
 B. II only
 C. III only
 D. I and II only
 E. I, II and III only

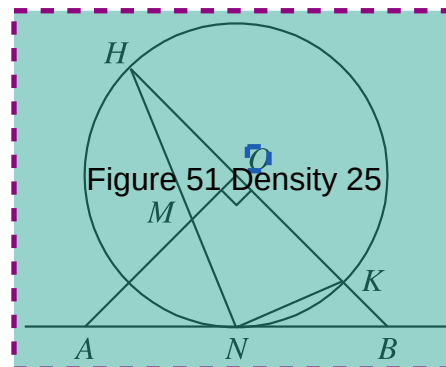
Label 50



In the figure, TP and TQ are tangent to the circle at P and Q respectively. If M is a point on the minor arc PQ and $\angle PMQ = \theta$, then $\angle PTQ =$

- A. $\frac{\theta}{2}$
 B. $\theta - 90^\circ$
 C. $180^\circ - \theta$
 D. $180^\circ - 2\theta$
 E. $2\theta - 180^\circ$

Label 51



In the figure, O is the centre of the circle. AB touches the circle at N . Which of the following is/are correct?

- I. M, N, K, O are concyclic.
 II. $\triangle HNB \sim \triangle NKB$
 III. $\angle OAN = \angle NOB$

- A. I only
 B. II only

- C. III only
- D. I and II only
- E. I, II and III only

Label 52
52

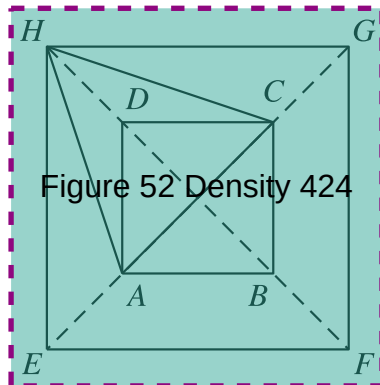


Figure 52 Density 424

In the figure $ABCD$ and $EFGH$ are two squares, and ACH is an equilateral triangle. Find $AB : EE$.

- A. 1 : 2
- B. 1 : 3
- C. $1 : \sqrt{2}$
- D. $1 : \sqrt{3}$
- E. $\sqrt{2} : \sqrt{3}$

Label 53
53

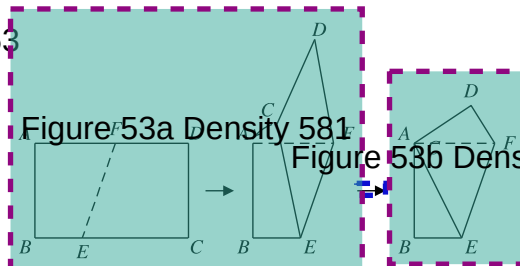


Figure 53a Density 581

Figure 53b Density 891

In the figure, a rectangular piece of paper $ABCD$ is folded along EF so that C and A coincide. If $AB = 12$ cm, $BC = 16$ cm, find BE .

- A. 3.5 cm
- B. 4.5 cm
- C. 5 cm
- D. 8 cm
- E. 12.5 cm

Label 54
54

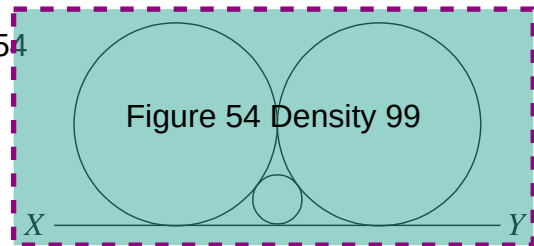


Figure 54 Density 99

In the figure, the three circles touch one another. XY is their common tangent. The two larger circles are equal. If the radius of the smaller circle is 4 cm, find the radii of the larger circles.

- A. 8 cm
- B. 10 cm
- C. 12 cm
- D. 14 cm
- E. 16 cm